

5 Appendix 2: AO-Cat Properties

Inverse properties are specified only for properties whose range is not a datatype.

5.1 has_identifier

Domain:	AO_Entity
Range:	xsd:string
Subproperty of:	
Superproperty of:	
Obligation:	Optional
Scope Note:	This property associates an ARIADNE entity with an identifier for that entity in some namespace other than the ARIADNE namespace (where the primary identifier of the entity belongs) and other than the provider's namespace (which is associated to data resources by property has_original_id). The range of this property is xsd:string for generality.
Maps to:	has_identifier is a shortcut of the fully developed path crm:P1_is_identified_by → crm:E42_Identifier
Hints:	values of dct:identifier can be used to compute the ARIADNE identifier.

5.2 has_type

Domain:	AO_Entity
Range:	AO_Concept
Subproperty of:	
Superproperty of:	
Inverse:	is_type_of
Obligation:	Mandatory
Scope Note:	This property associates an ARIADNE Entity with a term from a controlled vocabulary.
Example:	Event E has type "Rescue Excavation" (https://heritagedata.org/live/schemes/agl_et/concepts/145157.html)
Maps to:	has_type is equivalent to crm:P2_has_type
Hints:	this property is a sub-property of dct:type

5.3 has_title

Domain:	AO_Entity
Range:	rdfs:Literal
Subproperty of:	
Superproperty of:	
Obligation:	Mandatory with multiple instances (related to language, we may have one title for each relevant language). It is strongly suggested to use literals with a language tag for each provided title, such as "La bella addormentata"@it.

Scope Note: This property associates an AO Entity with one title identifying the resource in a specific language. A resource may have more than one title, but they must all be in different languages.

Maps to: has_title is a shortcut of the fully developed path
 crm:P102_has_title → crm:E35_Title

Hints: this property is a sub-property of dct:title

5.4 has_description

Domain: AO_Entity

Range: xsd:string

Subproperty of:

Superproperty of:

Obligation: Desirable

Scope Note: This property associates an ARIADNE entity with a free text description of that entity.

Maps to: This property is a subproperty of crm:P3_has_note

Hints: this property is a sub-property of dct:description and of rdfs:comment

5.5 was_issued

Domain: AO_Resource

Range: xsd:dateTime

Subproperty of:

Superproperty of:

Obligation: Mandatory

Scope Note: This property associates an ARIADNE resource with the date of formal issuance (e.g., publication) of the resource by the publisher.

Maps to: This property is a shortcut of the fully developed path
 →L10i→D7[10]→P4→E52→P81→XSD:DateTime
 →L10i→D7[10]→P2→E55["Issuance"]

5.6 was_modified

Domain: AO_Resource

Range: xsd:dateTime

Subproperty of:

Superproperty of:

Obligation: Mandatory

Scope Note: This property associates an ARIADNE resource with the most recent date on which the resource was modified by the publisher.

Maps to: It is assumed that the Modification is a new creation and the resource is the resulting output of the new Digital Machine Event. Therefore, this property is interpreted as a shortcut of the fully developed path
 →L11i_was_output_of→D7[11]_Digital_Machine_Event→P4_has_time-span→E52_Time-Span→P82_at_some_time_within→xsd:date
 →L11i_was_output_of→D7[11]_Digital_Machine_Event→P2_has_type→E55_Ty

pe["Modification"]

5.7 has_part

Domain:	AO_Collection
Range:	AO_Data_Resource
Subproperty of:	
Superproperty of:	
Inverse:	is_part_of
Obligation:	Desirable
Scope Note:	Associates an ARIADNE collection with a AO_Data_Resource resource that the collection contains. The relation between a collection in its members must be a tree having the collection as its root and any finite depth, due to the fact that a collection may have another collection as a member.
Maps to:	This property is equivalent to crmpe:PP20_has_persistent_dataset_part which is a subproperty of crm:P106_is_composed_of but it restricts the domain to Collections only. It doesn't apply for AO_Data_Resource and AO_Individual_Data_Resource.
Hints:	This property is a sub-property of dct:hasPart

5.8 has_publisher

Domain:	AO_Resource
Range:	AO_Agent
Subproperty of:	
Superproperty of:	
Inverse:	is_publisher_of
Obligation:	Mandatory
Scope Note:	Associates any ARIADNE resource with an agent responsible for making the resource publicly accessible (via download, or API, or other).
Maps to:	This property is a shortcut of the fully developed paths →L10i_was_input_of→D7[12]_Digital_Machine_event→P01i_is_domain_of→PC14_carried_out_by→P02_has_range→E39_Actor →L10i_was_input_of→D7[12]→P01i_is_domain_of→PC14_carried_out_by→P14.1_in_the_role_of→E55["Publisher"] →L10i_was_input_of→D7[12]_Digital_Machine_event→P2_has_type→E55_Type["Ariadne Content Provision"]
Hints:	This property is a sub-property of dct:publisher

5.9 has_contributor

Domain:	AO_Resource
Range:	AO_Agent
Subproperty of:	

Superproperty of:

Inverse: is_contributor_of

Obligation: Mandatory

Scope Note: Associates any ARIADNE resource with an agent primarily responsible for contributing the description of the resource to the ARIADNE Content Cloud.

Maps to: This property is a shortcut of the fully developed paths
→P94i_was_created_by→E65[13]Creation→P01i_is_domain_of→PC14_carried_out_by→P02_has_range→E39_Actor

→P94i_was_created_by→E65[13]Creation→P01i_is_domain_of→PC14_carried_out_by→P14.1_in_the_role_of→E55["Contributor"]

→P94i_was_created_by→E65[13]Creation→P2_has_type→E55_Type["Ariadne Content Creation"]

Hints: This property is a sub-property of dct:contributor

5.10 has_creator

Domain: AO_Resource

Range: AO_Agent

Subproperty of:

Superproperty of:

Inverse: is_creator_of

Obligation: Mandatory

Scope Note: Associates any ARIADNE resource with an agent primarily responsible for creating the resource.

Maps to: This property is a shortcut of the fully developed paths
→P94i_was_created_by→E65[13]Creation→P01i_is_domain_of→PC14_carried_out_by→P02_has_range→E39_Actor

→P94i_was_created_by→E65[13]Creation→P01i_is_domain_of→PC14_carried_out_by→P14.1_in_the_role_of→E55["Creator"]

→P94i_was_created_by→E65[13]Creation→P2_has_type→E55_Type["Ariadne Content Creation"]

Hints: This property is a sub-property of dct:creator

5.11 has_owner

Domain: AO_Resource

Range: AO_Agent

Subproperty of:

Superproperty of:

Inverse: is_owner_of

Obligation: Mandatory

Scope Note: Associates any ARIADNE resource with the legal owner of the resource, who holds the legal responsibility.

Maps to: This property is a shortcut of the fully developed paths
→P104_is_subject_to→E30_Right→P75i_is_posessed_by→E39_Actor
→P104_is_subject_to→E30_Right→P2_has_type→E55_Type["ownership"]

5.12 has_responsible

Domain: AO_Resource
Range: AO_Agent
Subproperty of:
Superproperty of:
Inverse: is_responsible_of
Obligation: Mandatory
Scope Note: Associates any ARIADNE resource with an agent holding the scientific responsibility of the resource, such as the person or team who had enough competence and creativity to conceive the service or to gather the data on a field.

Maps to: This property is a shortcut of the fully developed paths
→L10i_was_input_of→D7[12]_Digital_Machine_event→P01i_is_domain_of→PC14_carried_out_by→P02_has_range→E39_Actor

→L10i_was_input_of→D7[12]→P01i_is_domain_of→PC14_carried_out_by→P14.1_in_the_role_of→E55["Scientific or Technical Responsible"]

→L10i_was_input_of→D7[12]_Digital_Machine_event→P2_has_type→E55_Type["Ariadne Content Provision"]

5.13 has_visual_component

Domain: AO_Resource
Range: AO_Digital_Media
Subproperty of:
Superproperty of: has_primary_visual_component
Inverse: is_visual_component_of
Obligation: Optional
Scope Note: This property associates a resource with a digital media that gives some information about the resource.

Examples: (example given in Section [2.1 Adding visual items to AO-Cat descriptions](#))

Maps to: this property is a sub-property of crm:P106 is component of

Hints:

5.14 has_primary_visual_component

Domain: AO_Resource
Range: AO_Digital_Media
Subproperty of: has_visual_component

Superproperty of:

Inverse: is_primary_visual_component_of

Obligation: Optional

Scope Note: This property associates a resource with a digital media that gives some information about the resource and that has to be preferred over other media associated to the same resource whenever a single media is required.

Examples: (example given in Section [2.1 Adding visual items to AO-Cat descriptions](#))

Maps to:

Hints:

5.15 has_original_id

Domain: AO_Data_Resource

Range: xsd:string

Subproperty of:

Superproperty of:

Obligation: Mandatory

Scope Note: This property associates a data resource with the local identifier of the resource supplied by the content provider.

Examples:

Maps to: has_original_id is a shortcut of the fully developed path
crm:P1_is_identified_by → crm:E42_Identifier[1]→ crm:P2_has_type_→
crm:E55_Type["originalID"]

5.16 refers_to

Domain: AO_Data_Resource

Range: AO_Entity

Subproperty of:

Superproperty of: is_about

Inverse: is_referenced_by

Obligation: Optional

Scope Note: This property associates a AO_Data_Resource to the AO_Entity(ies) the resource refers to

Maps to: Subproperty of crm:P67_refers_to

Hints: This property is a sub-property of the inverse of dct:isReferencedBy

5.17 is_about

Domain: AO_Data_Resource

Range: AO_Entity

Subproperty of: refers_to

Superproperty of: has_ARIADNE_subject, has_native_subject, has_derived_subject

Inverse: is_subject_of

Obligation: Mandatory

Scope Note: This property associates a AO_Data_Resource to the AO_Entity the resource is about.

Maps to: Subproperty of crm:P129_is_about

Hints: This property is a sub-property of dc:subject

5.18 has_ARIADNE_subject

Domain: AO_Data_Resource

Range: AO_Concept

Subproperty of: is_about

Superproperty of:

Inverse: is_ARIADNE_subject_of

Obligation: Mandatory

Scope Note: This property associates a AO_Data_Resource with one the fundamental archaeological categories defined by ARIADNE. These are the high level “resource types”, or semantic categories which are used in the ARIADNE portal to filter search results, e.g. “Site and monuments databases or inventories”, “Event/intervention resources” *etc.*

Maps to: has_ARIADNE_subject is a shortcut of the fully developed path
→P129_is_about→E55_Type→P2_has_type→E55_Type["Ariadne Subject"]

Hints: This property is a sub-property of dc:subject

5.19 has_native_subject

Domain: AO_Data_Resource

Range: AO_Concept

Subproperty of: is_about

Superproperty of:

Inverse: is_native_subject_of

Obligation: Mandatory

Scope Note: This property associates a AO_Data_Resource with an original subject in the providing institution

Maps to: has_native_subject is a shortcut of the fully developed path
→P129_is_about→E55_Type→P2_has_type→E55_Type["Native Subject"]

Hints: This property is a sub-property of dc:subject

5.20 has_derived_subject

Domain: AO_Data_Resource

Range: AO_Concept (Getty AAT Term)

Subproperty of: is_about

Superproperty of:

Inverse: is_derived_subject_of

Obligation: Mandatory (automatically computed by existing mapping to AAT)

Scope Note: This property associates a AO_Data_Resource with a subject automatically derived by mapping a native subject of the resource to a

term in the Getty AAT vocabulary. The native subject may have the SKOS relations (skos:broadMatch, skos:closeMatch, skos:exactMatch, skos:narrowMatch, skos:relatedMatch) to the Getty AAT vocabulary.

Maps to: has_derived_subject is a shortcut of the fully developed path
→P129_is_about→E55_Type→P2_has_type→E55_Type["Derived Subject"]

Hints: This property is a sub-property of dc:subject

5.21 has_language

Domain: AO_Data_Resource

Range: AO_Concept (linked to lexvo.org)

Subproperty of:

Superproperty of:

Inverse: is_language_of

Obligation: Optional

Scope Note: This property is used to associate a data resource with the language of the resource, if any, specified according to a selected vocabulary.

Maps to: Equivalent to crm:P72_has_language

Hints: This property is a sub-property of dc:language and its range is a super-class of class Ivont:Language

5.22 was_created_on

Domain: AO_Data_Resource

Range: xsd:dateTime

Subproperty of:

Superproperty of:

Obligation: Mandatory

Scope Note: This property is used to specify the creation date of the AO_Data_Resource. This is the date when the AO_Data_Resource was first made available online by the Publisher, (not the date of the fieldwork or laboratory analysis for instance). In some cases (e.g. where data or metadata did not already exist online) it will be the same date as the creation of the metadata for ARIADNE

Maps to: This property is a shortcut of the fully developed path
→P94i_was_created_by→E65_Creation→P4_has_time-span→E52_Time-Span
→P81_ongoing_throughout→xsd:date

Hints: This property is a sub-property of dct:created (and consequently of dc:date)

5.23 has_landing_page

Domain: AO_Data_Resource

Range: rdfs:Resource

Subproperty of:
 Superproperty of:
 Obligation: Optional
 Scope Note: This property is used to specify the original landing page of the AO_Data_Resource, if any.
 Examples: The Archaeology Data Service has landing page e.g. <http://archaeologydataservice.ac.uk/archsearch/record.jsf?titleId=104031> e.g. <https://doi.org/10.5284/1052661>
 Maps to: This property is a shortcut of the fully developed path
 →PP8i_is_dataset_hosted_by→PE15_Data_E-Service→PP49_provides_access_point→PE29_Access_Point

5.24 has_access_policy

Domain: AO_Data_Resource
 Range: rdfs:Resource
 Subproperty of:
 Superproperty of:
 Obligation: Desirable
 Scope Note: This property is used to specify the URI to the statement of policy (typically, on an organization's website) for the data resource.
 Examples: The DANS Organization has access policy as stated in <http://dans.knaw.nl/en/about/organisation-and-policy/legal-information>
 Maps to: This property is a shortcut of the fully developed path
 →P104_is_subject_to→E30_Right→P2_has_type→E55_Type["Access Policy"]

5.25 has_access_rights

Domain: AO_Data_Resource
 Range: xsd:string
 Subproperty of:
 Superproperty of:
 Obligation: Mandatory
 Scope Note: This property contains information about who can access the resource or an indication of its security status.
 Examples: The data resource xyz has access rights "CC BY-NC-SA 2.5 SI"
 Maps to: This property is a shortcut of the fully developed path
 →P104_is_subject_to→E30_Right→P2_has_type→E55_Type["Access Rights"]
 Hints: This property is a sub-property of dct:accessRights

5.26 has_extent

Domain: AO_Data_Resource
 Range: xsd:string
 Subproperty of:
 Superproperty of:

Obligation: Optional
 Scope Note: This property specifies the size of the AO_Data_Resource (i.e., number of members in a collection, number of records in a dataset, etc.).
 Examples: Collection X has extent [100 item]
 Maps to: Equivalent to crm:P43_has_dimension
 Hints: This property is a sub-property of dct:extent

5.27 has_temporal_coverage

Domain: AO_Data_Resource
 Range: AO_Temporal_Region
 Subproperty of:
 Superproperty of:
 Inverse: is_temporal_coverage_of
 Obligation: Desirable
 Scope Note: This property associates a data resource with the temporal region covered by the content of the resource.
 Maps to: Equivalent to crm:P129_is_about with a restricted range E4_Period
 Hints: This is a sub-property of dct:temporal

5.28 occurs_in

Domain: AO_Event
 Range: AO_Spatial_Region
 Subproperty of:
 Superproperty of:
 Inverse: is_spatial_region_of
 Obligation: Mandatory
 Scope Note: This property associates an ARIADNE Event with the spatial region in which the event occurred.
 Maps to: Equivalent to crm:P7_took_place_at

5.29 happens_during

Domain: AO_Event
 Range: AO_Temporal_Region
 Subproperty of:
 Superproperty of:
 Inverse: is_temporal_region_of
 Obligation: Desirable
 Scope Note: This property associates an ARIADNE Event with the temporal region at which the event occurred.
 Maps to: Equivalent to crm:P4_has_time-span

5.30 contains_event

Domain: AO_Event
Range: AO_Event
Subproperty of:
Superproperty of:
Inverse: is_contained_in
Obligation: Optional
Scope Note: This property associates an ARIADNE Event with another ARIADNE Event that is part of it. As a consequence, the former event occurred in a spatio-temporal region that is contained in the spatio-temporal region of the latter event.
Maps to: Equivalent to crm:P9_consists_of

5.31 has_period

Domain: AO_Temporal_Region
Range: AO_Concept (from PeriodO)
Subproperty of:
Superproperty of:
Inverse: is_period_of
Obligation: Desirable
Scope Note: This property associates a AO_Temporal_Region with a temporal period defined in periodO, so it ranges over web resources.
Maps to: crm:P2_has_type->E55_Type->P2_has_type->E55_Type["PeriodO"]

5.32 has_native_period

Domain: AO_Temporal_Region
Range: AO_Concept
Subproperty of:
Superproperty of:
Inverse: is_native_period_of
Obligation: Mandatory
Scope Note: This property associates a AO_Temporal_Region with a concept representing a period in some local vocabulary of the provider. Its range is therefore AO_Concept.
Examples:
Maps to: crm:P2_has_type->E55_Type->P2_has_type->E55_Type["Native Period"]

5.33 has_spatial_coverage

Domain: AO_Data_Resource
Range: AO_Spatial_Region

Subproperty of:
 Superproperty of:
 Inverse: is_spatial_coverage_of
 Obligation: Mandatory
 Scope Note: This property associates a data resource with the spatial region covered by the content of the resource.
 Maps to: Equivalent to crm:P129_is_about with a restricted range E53_Place
 Hints: This is a sub-property of dc:spatial

5.34 has_spatial_precision

Domain: AO_Data_Resource
 Range: AO_Dimension
 Subproperty of:
 Superproperty of:
 Inverse: is_spatial_precision_of
 Obligation: Optional
 Scope Note: This property associates a data resource with the approximation with which the spatial region covered by the content of the resource is specified.
 Maps to: Sub-property of crm:P43_has_dimension

5.35 has_value

Domain: AO_Dimension
 Range: xsd:decimal
 Subproperty of:
 Superproperty of:
 Inverse:
 Obligation: Optional
 Scope Note: This property associates a numeric value with a dimension.
 Maps to: Sub-property of crm:P90_has_value

5.36 has_unit

Domain: AO_Dimension
 Range: AO_Concept
 Subproperty of:
 Superproperty of:
 Inverse:
 Obligation: Optional
 Scope Note: This property associates a unit of measure with a dimension.
 Maps to: Sub-property of crm:P2_has_note

5.37 is_accessible_at

Domain:	AO_Service
Range:	rdfs:Resource
Subproperty of:	
Superproperty of:	
Inverse:	is_access_point_of
Obligation:	Mandatory
Scope Note:	This property is used to associate a service with an IRI where the service is accessible. If the service is a web service, this IRI is the technical access point of the service. Otherwise, it is the IRI of a resource describing how the service can be accessed.
Examples:	The ARIADNE Visual service is accessible at http://visual.ariadne-infrastructure.eu/
Maps to:	Equivalent to crmpe:PP28_has_designated_access_point

5.38 has_functionality

Domain:	AO_Service
Range:	AO_Concept
Subproperty of:	
Superproperty of:	
Inverse:	is_functionality_of
Obligation:	Mandatory
Scope Note:	This property associates a service with its functionality, expressed in some vocabulary.
Examples:	The ARIADNE visual service has functionality “3D visualization”
Maps to:	This property is equivalent to crmpe:PP45_has_competency

5.39 has_consumed_media

Domain:	AO_Service
Range:	AO_Concept
Subproperty of:	
Superproperty of:	
Inverse:	is_consumed_media_of
Obligation:	Mandatory
Scope Note:	This property associates a service with the media type(s) handled by the service. The list of possible mediaType is open.
Examples:	The ARIADNE visual service has consumed media “image”, “2D model”
Maps to:	Equivalent to crm:P125_used_object_of_type

5.40 has_produced_media

Domain:	AO_Service
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Range: AO_Concept
 Subproperty of:
 Superproperty of:
 Inverse: is_produced_media_of
 Obligation: Mandatory
 Scope Note: This property specifies the media types of the objects created by the service. The list of possible mediaType is open.
 Scope Note: This property associates a service with the media type(s) produced by the service. The list of possible mediaType is open.
 Examples: "image", "audio", "video", "text", "3D model", "2D model"
 Maps to: This property is a shortcut of the fully developed path
 crmdig:L11_had_output->crmdig:D1_Digital_Object->crm:P2_has_type->crm:E55_Type->crm:P2_has_type->crm:E55_Type["media"]

5.41 has_consumed_format

Domain: AO_Service
 Range: AO_Concept (the IRI of a MIME type)
 Subproperty of:
 Superproperty of:
 Inverse: is_the_format_consumed_by
 Obligation: Mandatory
 Scope Note: This property specifies the MIME type of the objects handled by the service.
 Examples: "iana:image/jpeg"
 Maps to: This property is a shortcut of the fully developed path
 crmdig:L10_had_input->crmdig:D1_Digital_Object->crm:P2_has_type->crm:E55_Type->crm:P2_has_type->crm:E55_Type["format"]

5.42 has_produced_format

Domain: AO_Service
 Range: AO_Concept (the IRI of a MIME type)
 Subproperty of:
 Superproperty of:
 Inverse: is_the_format_produced_by
 Obligation: Mandatory
 Scope Note: This property specifies the MIME type of the objects created by the service.
 Examples: "iana:image/jpeg"
 Maps to: This property is a shortcut of the fully developed path
 crmdig:L11_had_output->crmdig:D1_Digital_Object->crm:P2_has_type->crm:E55_Type->crm:P2_has_type->crm:E55_Type["format"]

5.43 has_supported_language

Domain: AO_Service
Range: AO_Concept (linked to lexvo.org)
Subproperty of:
Superproperty of:
Obligation: Desirable
Scope Note: This property specifies the languages supported by the service, encoded according with ISO 639 standard (ISO 639-1:2002).
Examples: The Pisa Meteo service has supported language “it”
Maps to: Equivalent to crm:P72_has_language
Hints: This property is a sub-property of dc:language

5.44 has_technical_support

Domain: AO_Service
Range: AO_Agent
Subproperty of:
Superproperty of:
Inverse: is_technical_support_of
Obligation: Mandatory
Scope Note: This property specifies the agent offering technical support on the service, if any.
Examples: AO_Service C has_technical_support <https://www.isti.cnr.it/>
Maps to: Subproperty of crmpe:PP2_provided_by

5.45 has_time_interval

Domain: AO_Object
Range: AO_Temporal_Region
Subproperty of:
Superproperty of:
Inverse: is_time_interval_of
Obligation: Optional
Scope Note: This property connects an object to a temporal region relevant for that object
Examples:
Maps to: This property is a shortcut of the fully developed path
crm:P140i_was_attributed_by->
crm:E13_Attribute_Assignment->crm:P141_assigned->
crm:E52_Time-Span

5.46 has_space_region

Domain: AO_Object
Range: AO_Spatial_Region
Subproperty of:
Superproperty of:
Inverse: is_space_region_of
Obligation: Optional
Scope Note: This property connects an object to a spatial region relevant for that object
Examples:
Maps to: This property is a subproperty of crm:P53_has_former_or_current_location

5.47 was_present_at

Domain: AO_Object
Range: AO_Event
Subproperty of:
Superproperty of:
Inverse: occurred_in_the_presence_of
Obligation: Optional
Scope Note: This property connects an object to an event in which the object plays a relevant role
Examples:
Maps to: This property is a subproperty of crm:P12i_was_present_at

5.48 has_name

Domain: AO_Agent
Range: xsd:string
Subproperty of:
Superproperty of:
Obligation: Mandatory
Scope Note: This property is used to specify the name of the person, institution or organisation related with an AO_Resource
Examples: Julian Richards has name "Julian Richards"
Maps to: This property is a shortcut of the fully developed path
→ P1_is_identified_by → E41_Appellation → P2_has_type → E55_Type['Name']
Hints: This property is a sub-property of foaf:name

5.49 has_agent_identifier

Domain: AO_Agent

Range: xsd:string
 Subproperty of:
 Superproperty of:
 Obligation: Desirable
 Scope Note: This property is used to specify the identifier of the person, institution or organisation related with an AO_Resource. The identifier also specifies the authority that provides it.
 Examples: M. Theodoridou has ORCID "http://orcid.org/0000-0002-4623-9186"
 Maps to: This property is a shortcut of the fully developed path
 → P1_is_identified_by→ E41_Appellation→ P2_has_type →
 E55_Type[Authority]
 Hints: This property is a sub-property of foaf:name

5.50 has_email

Domain: AO_Agent
 Range: xsd:string
 Subproperty of:
 Superproperty of:
 Obligation: Desirable
 Scope Note: This property is used to specify the mail address of the person, institution or organisation related with an AO_Resource
 Examples: name@institution.org
 Maps to: This property is a shortcut of the fully developed path
 crm:P76_has_contact_point->E51_Contact_Point->P2_has_type->E55_Type["Email"]
 Hints: This property is a sub-property of foaf:mbox

5.51 has_homepage

Domain: AO_Agent
 Range: rdfs:Resource
 Subproperty of:
 Superproperty of:
 Inverse: is_homepage_of
 Obligation: Optional
 Scope Note: This property is used to specify the website home page of the person, institution or organisation related with an AO_Resource
 Examples: http://www.vast-lab.org
 Maps to: This property is a shortcut of the fully developed path
 crm:P76_has_contact_point->E51_Contact_Point->P2_has_type->E55_Type["homepage"]
 Hints: This property is a sub-property of foaf:homepage

5.52 from

Domain: AO_Temporal_Region
Range: xsd:gYear
Subproperty of:
Superproperty of:
Obligation: Desirable
Scope Note: This property is used to indicate the starting date of a AO_Temporal_Region in ISO 8601 format.
Examples: Temporal region X starts from 312
Maps to: This property is equivalent to crm:P79_beginning_is_qualified_by

5.53 until

Domain: AO_Temporal_Region
Range: xsd:gYear
Subproperty of:
Superproperty of:
Obligation: Desirable
Scope Note: This property is used to indicate the ending date of a AO_Temporal_Region in ISO 8601 format.
Examples: Temporal region X lasts until 1491
Maps to: This property is equivalent to crm:P80_end_is_qualified_by

5.54 has_place_name

Domain: AO_Spatial_Region
Range: xsd:string
Subproperty of:
Superproperty of:
Obligation: Mandatory
Scope Note: This property is used to identify a SpatialRegion with its place name.
Examples: Spatial region Z has place name "Florence"
Maps to: This property is a sub-property of P87_is_identified_by

5.55 has_coordinate_system

Domain: AO_Spatial_Region
Range: xsd:string
Subproperty of:
Superproperty of:
Obligation: Optional
Scope Note: This property indicates the coordinate system used to encode coordinates.
Examples: Spatial region W has coordinate system "EPSG 2763"

Maps to: This property is a shortcut of the fully developed path
→ P87_is_identified_by → E47_Spatial_Coordinates → P2_has_type
→ E55_Type[*coordinate-system-name*]

5.56 has_country

Domain: AO_Spatial_Region
Range: rdfs:Resource
Obligation: Mandatory
Scope Note: This property indicates the country to which a spatial region belongs. The range is the identifier of the country (such as a Wikidata URI) specified by instantiating an rdfs:Resource. The corresponding country name can be provided by associating an rdfs:label with the instantiated rdf:Resource.
Examples: Spatial region W has country Afghanistan (Wikidata URI: <https://www.wikidata.org/entity/Q889>).

```
<rdf:Description rdf:about="https://[...]/AO_Spatial_Region_W">  
  <aocat:has_country rdf:resource="http://www.wikidata.org/entity/Q889"/>  
</rdf:Description>
```

```
<rdf:Description rdf:about="https://www.wikidata.org/entity/Q889">  
  <rdfs:label xml:lang="en">Afghanistan</rdfs:label>  
</rdf:Description>
```

Maps to: This property is a subproperty of P3 has note.

5.57 has_latitude

Domain: AO_Spatial_Region_Point
Range: xsd:decimal
Subproperty of:
Superproperty of:
Obligation: Desirable
Scope Note: This property indicates the latitude value of the coordinates of a AO_Spatial_Region.
Examples: Spatial region W has latitude 54.540575
Maps to: This property is a shortcut of the fully developed path
→ P87_is_identified_by → E47_Spatial_Coordinates → P2_has_type
→ E55_Type["latitude"]
Hints: This is a sub-property of geo:lat

5.58 has_longitude

Domain:	AO_Spatial_Region_Point
Range:	xsd:decimal
Subproperty of:	
Superproperty of:	
Obligation:	Desirable
Scope Note:	This property indicates the minimum longitude value of the coordinates of a AO_Spatial_Region.
Examples:	Spatial region W has longitude -3.293877
Maps to:	This property is a shortcut of the fully developed path → P87_is_identified_by → E47_Spatial_Coordinates → P2_has_type → E55_Type["longitude"]
Hints:	This is a sub-property of geo:lon

5.59 has_bounding_box_min_lat

Domain:	AO_Spatial_Region_BBox
Range:	xsd:decimal
Subproperty of:	
Superproperty of:	
Obligation:	Optional
Scope Note:	This property indicates the minimum latitude value of the bounding box area defining a AO_Spatial_Region.
Examples:	Spatial region W has bounding box min lat 49.85215377807617
Maps to:	This property is a shortcut of the fully developed path → P87_is_identified_by → E47_Spatial_Coordinates → P2_has_type → E55_Type["boundingBoxMinLat"]

5.60 has_bounding_box_min_lon

Domain:	AO_Spatial_Region_BBox
Range:	xsd:decimal
Subproperty of:	
Superproperty of:	
Obligation:	Optional
Scope Note:	This property indicates the minimum longitude value of the bounding box area defining a AO_Spatial_Region.
Examples:	Spatial region W has bounding box min lon -6.0230712890625
Maps to:	This property is a shortcut of the fully developed path → P87_is_identified_by → E47_Spatial_Coordinates → P2_has_type → E55_Type["boundingBoxMinLon"]

5.61 has_bounding_box_max_lat

Domain: AO_Spatial_Region_BBox
Range: xsd:decimal
Subproperty of:
Superproperty of:
Obligation: Optional
Scope Note: This property indicates the maximum latitude value of the bounding box area defining a AO_Spatial_Region.
Examples: Spatial region W has bounding box max lat 55.7147331237793
Maps to: This property is a shortcut of the fully developed path
→ P87_is_identified_by → E47_Spatial_Coordinates → P2_has_type
→ E55_Type["boundingBoxMaxLat"]

5.62 has_bounding_box_max_lon

Domain: AO_Spatial_Region_BBox
Range: xsd:decimal
Subproperty of:
Superproperty of:
Obligation: Optional
Scope Note: This property indicates the maximum longitude value of the bounding box area defining a AO_Spatial_Region.
Examples: Spatial region W has bounding box max lon 2.0408935546875
Maps to: This property is a shortcut of the fully developed path
→ P87_is_identified_by → E47_Spatial_Coordinates → P2_has_type
→ E55_Type["boundingBoxMaxLon"]

5.63 has_place_IRI

Domain: AO_Spatial_Region_StdName
Range: AO_Concept
Subproperty of:
Superproperty of:
Inverse: is_place_IRI_of
Obligation: Desirable (the gazetteer should be specified too)
Scope Note: This property associates a spatial region with a URI/IRI or other unique identifier from a standard gazetteer (e.g. Geonames for modern places, Pleiades for ancient places) used to refer a place or a spatialRegion.
Examples: Spatial region W has place IRI <http://www.geonames.org/3176959>
Maps to: This property is a shortcut of the fully developed path
→ P87_is_identified_by → E44_Place_Appellation → P2_has_type → E55_Type["gazetteer name"]

5.64 has_polygonal_representation

Domain: AO_Spatial_Region_Polygon
Range: XMLLiteral
Subproperty of:
Superproperty of:
Obligation: Optional
Scope Note: This property associates a spatial region with an XML representation of the polygon, typically handled by a GIS.
Examples:
Maps to: This property is a shortcut of the fully developed path
→ P87_is_identified_by → E47_Spatial_Coordinates → P2_has_type
→ E55_Type["polygon"]

5.65 has_institution

Domain: AO_Person
Range: AO_Concept
Subproperty of:
Superproperty of:
Obligation: Optional
Scope Note: This property specifies the institution where an agent belongs.
Examples: Agent Carlo has institution "ISTI"
Maps to:
Hints: This property is a sub-property of the inverse of foaf:member

5.66 is_mirror_of

Domain: AO_Digital_Media
Range: rdfs:Resource
Subproperty of:
Superproperty of:
Obligation: Optional
Scope Note: This property associates a digital media with the media it is a mirror of. The latter media is supposed to be on some provider's site, identified by an IRI.
Examples: (example given in [4.10.1 Digital images](#))
Maps to: This property is a shortcut of the fully developed path
→ PP8i_is_dataset_hosted_by → PE15_Data_E-Service → PP49_provides_access_point → PE29_Access_Point
Hints: this property is a sub-property of owl:sameAs

5.67 has_data_type

Domain:	AO_Individual_Data_Resource
Range:	AO_Concept
Subproperty of:	has_type
Inverse:	is_data_type_of
Obligation:	Mandatory
Scope Note:	This property associates an ARIADNE data resource with a term from a controlled vocabulary that specifies the data type of the resource. The data type refers to the categorisation of data based on its form, such as text, image, audio, video, etc. The terms used to describe the data type are defined in a controlled vocabulary of resource types specific to ARIADNE (e.g, "3D", "Image", "Audio", etc.).
Example:	The data resource xyz has_data_type "3D".
Maps to:	has_data_type is a subproperty of crm:P2_has_type.
Hints:	This property is a sub-property of dct:type.

5.68 has_data_format

Domain:	AO_Individual_Data_Resource
Range:	AO_Concept
Subproperty of:	has_type
Inverse:	is_data_format_of
Obligation:	Optional
Scope Note:	This property associates an ARIADNE data resource with a term from a controlled vocabulary that specifies the data format of the resource. The data format refers to the specific file format or encoding format used to store or represent the data, such as CSV, JSON, XML, JPEG, MP3, etc. The term will be selected from the MIME type, a standard vocabulary of formats providing a shared way of identifying the format of a resource on the internet. For example, the MIME type for a CSV file is "text/csv", and the MIME type for a JPEG file is "image/jpeg". The MIME type can be also used to help determine how to process or display the data resource.
Example:	The data resource xyz has data format "text/csv".
Maps to:	has_data_format is a subproperty of crm:P2_has_type
Hints:	This property is a sub-property of dct:type and equivalent to the encodingFormat property in Schema.org, which specifies the encoding format of a CreativeWork.

5.69 is_rendered_by

Domain: AO_Digital_Media

Range: AO_Service

Subproperty of:

Superproperty of:

Inverse: renders

Obligation: Optional

Scope Note: This property associates a digital media with the service that can render it. The service is identified by instantiating the AO_Service class. The name of the service can be provided by associating an rdfs:label with the instantiated AO_Service.

Examples:

Maps to:

Hints: